Attention Is All You Need

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The Challenge of Sequential Models

Dominant models like RNNs and LSTMs are inherently sequential, processing data token-by-token.

This sequential nature (h_t depends on h_{t-1}) prevents parallelization within examples.

Learning long-range dependencies is a key challenge due to long signal paths and vanishing gradients.

Memory constraints limit batching across examples for long sequences.

A New Architecture: The Transformer

We propose a network architecture based solely on attention mechanisms.

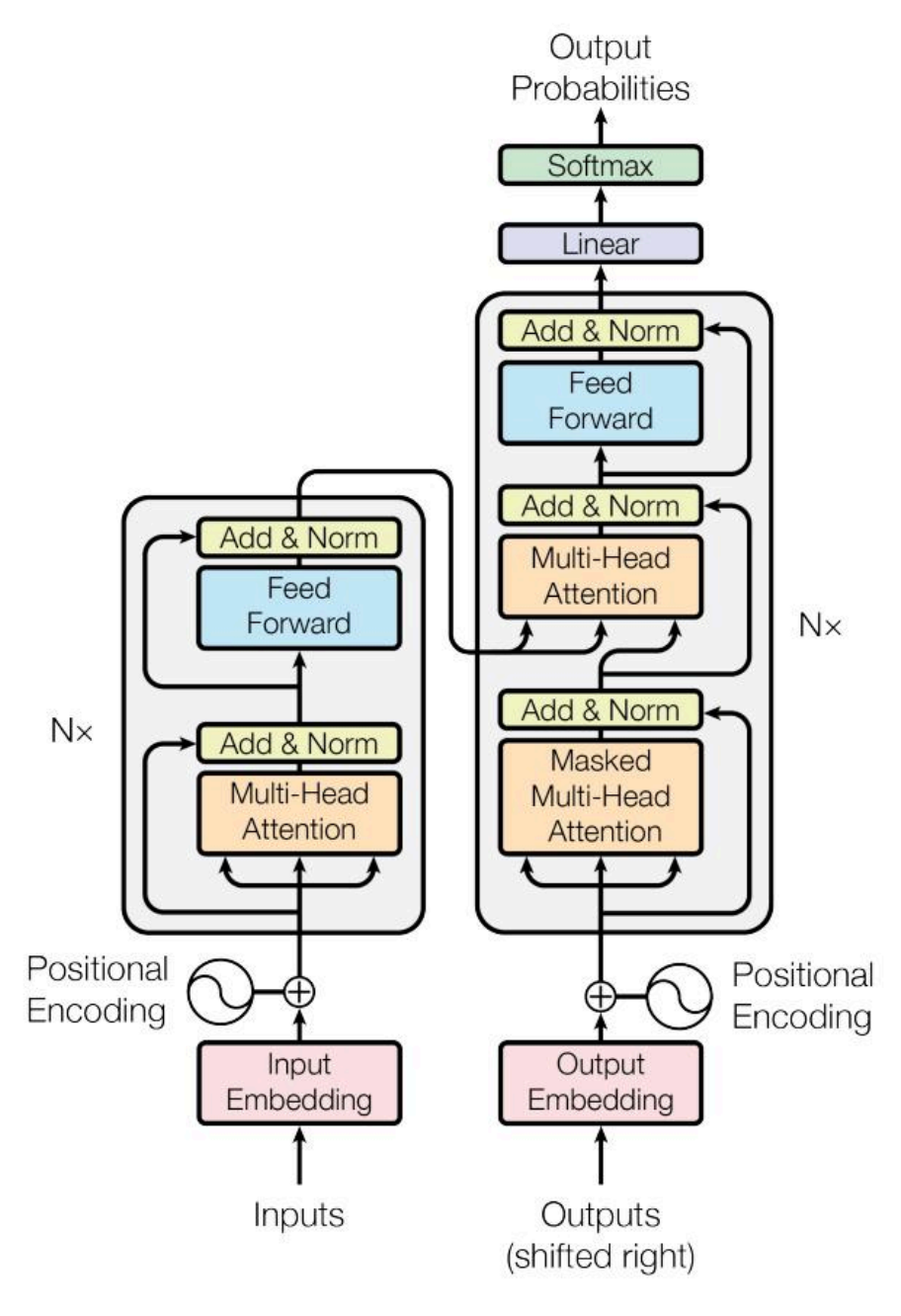
It dispenses with recurrence and convolutions entirely.

This design allows for significantly more parallelization, reducing training time.

Achieves state-of-the-art translation quality while requiring less time to train.

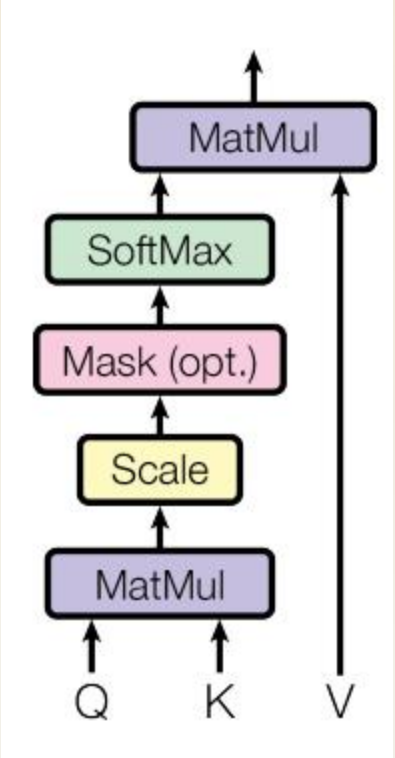
High-Level Architecture

- The Transformer follows a standard encoder-decoder structure.
- **Encoder:** Maps an input sequence of symbols to a sequence of continuous representations.
- **Decoder:** Generates an output sequence one symbol at a time, consuming previously generated symbols.
- Both components use stacked self-attention and point-wise, fully connected layers.



The Core Mechanism: Attention

- An attention function maps a query and a set of key-value pairs to an output.
- The output is a weighted sum of the values, where weights are computed based on query-key compatibility.
- We use a specific implementation called Scaled Dot-Product Attention.



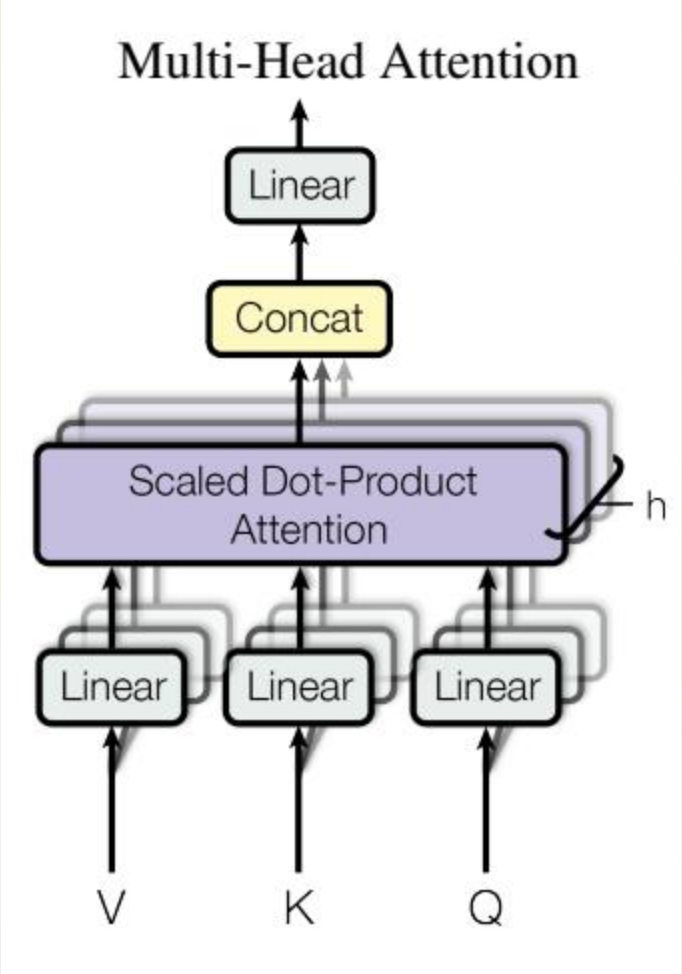
Scaled Dot-Product Attention

- Input consists of queries (Q), keys (K) of dimension d_k, and values (V) of dimension d_v.
- Dot products of the query with all keys are computed, then scaled by $\frac{1}{\sqrt{d_k}}$.
- Scaling prevents the softmax function from entering regions with small gradients, stabilizing training.
- A softmax is applied to obtain weights on the values.

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$$

Multi-Head Attention

- Instead of a single attention function, we use multiple attention 'heads' in parallel.
- Queries, keys, and values are linearly projected h times with different learned projections.
- This allows the model to jointly attend to information from different representation subspaces at different positions.
- Outputs from each head are concatenated and projected to form the final result.



Multi-Head Attention in Detail

- We use $h = 8$ parallel attention heads.
- For each head, we use reduced key, value, and query dimensions: $d_k = d_v = d_{model}/h = 64$.
- The total computational cost is similar to single-head attention with full dimensionality.

$$\begin{aligned} \mathrm{MultiHead}(Q, K, V) &= \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h) W^O \\ \text{where } \mathrm{head}_i &= \mathrm{Attention}(QW_i^Q, KW_i^K, VW_i^V) \end{aligned}$$

Layer Structure

Encoder Layer

- Composed of two sub-layers:
  - i. Multi-Head Self-Attention
  - ii. Position-wise Feed-Forward Network
- Residual connections and Layer Normalization are applied around each sub-layer.

Decoder Layer

- Composed of three sub-layers:
  - i. Masked Multi-Head Self-Attention
  - ii. Encoder-Decoder Attention
  - iii. Position-wise Feed-Forward Network
- Masking in the self-attention layer preserves the auto-regressive property.

Positional Encoding

- Since the model contains no recurrence or convolution, we must inject information about token position.
- We add 'positional encodings' to the input embeddings at the bottom of the encoder and decoder stacks.
- We use sine and cosine functions of different frequencies.

$$PE_{(pos,2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$$
$$PE_{(pos,2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$$

Why Self-Attention?

- Self-attention layers offer advantages in computational complexity, parallelization, and learning long-range dependencies.
- The path length between any two positions is constant ($O(1)$), unlike in recurrent ($O(n)$) or convolutional ($O(\log_k(n))$) layers.

LAYER TYPE	COMPLEXITY PER LAYER	SEQUENTIAL OPERATIONS	MAXIMUM PATH LENGTH
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
Self-Attention (restricted)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$

State-of-the-Art Machine Translation

- The Transformer outperforms the best previously reported models on WMT 2014 translation tasks.
- The big model establishes a new SOTA BLEU score of 28.4 on English-to-German.
- This performance is achieved at a fraction of the training cost of competing models.

MODEL	BLEU (EN-DE)	BLEU (EN-FR)	TRAINING COST (FLOPS)
GNMT + RL [38]	24.6	39.92	2.3×10^{19}
ConvS2S [9]	25.16	40.46	9.6×10^{18}
GNMT + RL Ensemble [38]	26.30	41.16	1.8×10^{20}
Transformer (base)	**27.3**	**38.1**	**3.3×10^{18}**
Transformer (big)	**28.4**	**41.8**	**2.3×10^{19}**

Ablation Studies

- We varied components to measure their importance on English-to-German translation.
- (A) 8 attention heads were found to be optimal.
- (D) Dropout is very helpful in avoiding over-fitting.
- (E) Sinusoidal positional encodings perform nearly identically to learned embeddings.

VARIATION	PPL (DEV)	BLEU (DEV)
base (h=8)	**4.92**	**25.8**
(A) h=1	5.29	24.9
(A) h=16	4.91	25.8
(D) P_{drop}=0.0	5.77	24.6
(E) Learned Positional Embedding	4.92	25.7
big	**4.33**	**26.4**

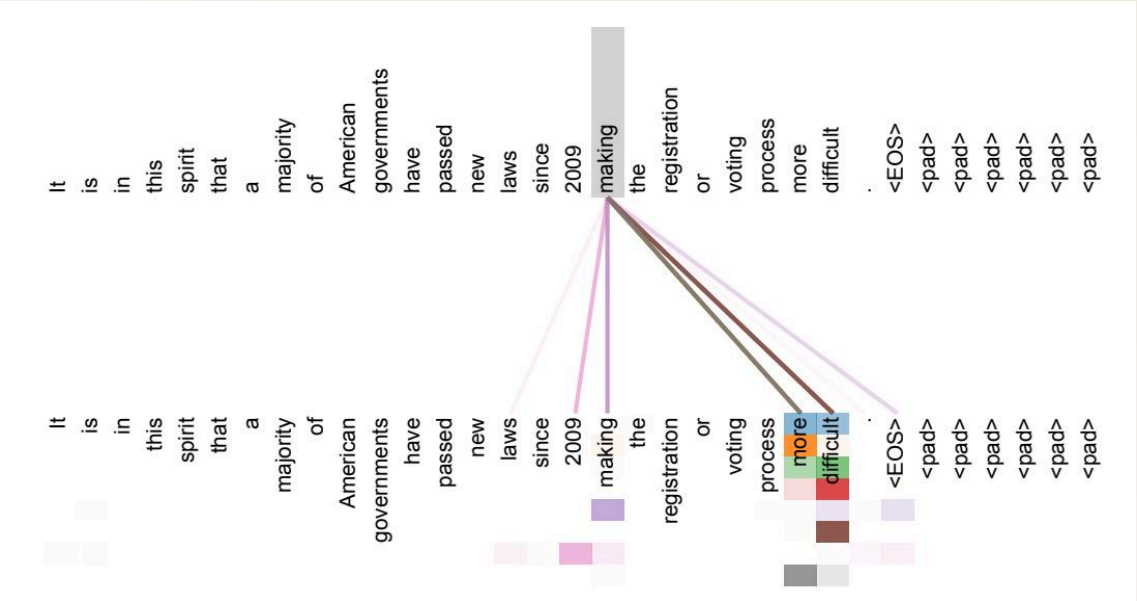
Generalizing to Other Tasks: Parsing

- To test generalization, we applied the Transformer to English constituency parsing.
- The model performs surprisingly well with minimal task-specific tuning.
- It outperforms the BerkeleyParser even when trained only on the small 40K sentence WSJ dataset.

PARSER	TRAINING SETTING	WSJ 23 F1 SCORE
Recurrent Neural Network Grammar	generative	93.3
Vinyals & Kaiser et al. (2015)	semi-supervised	92.1
Transformer (4 layers)	**semi-supervised**	**92.7**
Dyer et al. (2016)	WSJ only	91.7
Transformer (4 layers)	**WSJ only**	**91.3**

Visualizing Attention

- Self-attention mechanisms can lead to more interpretable models.
- We can inspect attention distributions to see how the model represents syntactic and semantic structures.
- Different attention heads learn to perform different tasks, such as tracking long-distance dependencies or resolving anaphora.



Conclusion

- We presented the Transformer, the first sequence transduction model based entirely on attention.
- It trains significantly faster and achieves a new state of the art on machine translation tasks.
- We are excited about the future of attention-based models and plan to apply them to other tasks and modalities.